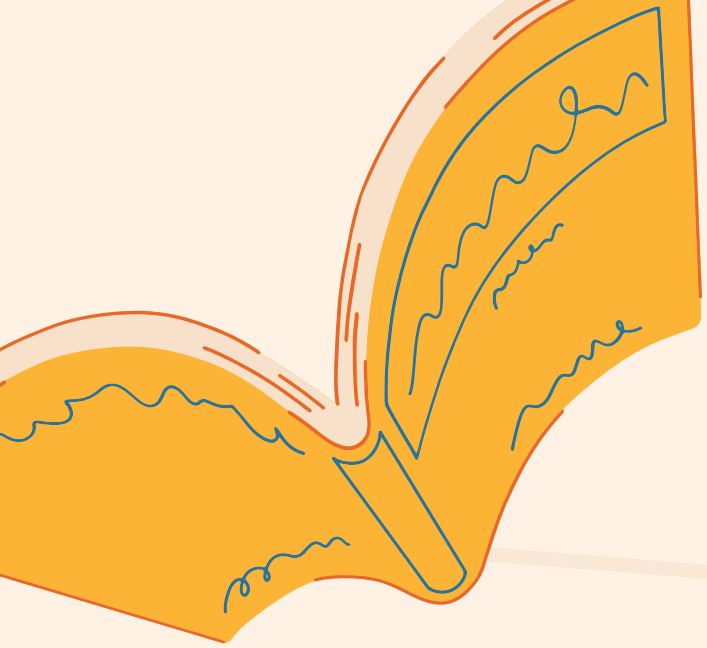




THE EFFECT OF NOTE-TAKING AND LEARNING STYLES TO COMPREHENSION AND MEMORY RECALL OF THE STUDENTS



PRESENTED BY:
DATA THE EXPLORER



INTRODUCTION

Learning styles – auditory, visual, and reading; affect how students process information, but note-taking remains a key tool for improving understanding and memory. Its effectiveness varies by learning style, highlighting the need to understand how different learners can use note-taking strategies to enhance comprehension and recall.



IMPORTANCE OF THE EXPERIMENT:

- Determines whether note-taking enhances comprehension and memory for different learning styles
- Guides teachers in creating more personalized and effective learning approaches
- Helps students adopt study habits suited to their strengths
- Improves academic performance in classrooms with diverse learners



RESEARCH OBJECTIVE(S):

- To determine the effect of note-taking on students' comprehension and memory recall.
- To identify which learning style; auditory, visual, or reading, shows the highest level of comprehension and memory recall.
- To examine the interaction effect between note-taking and learning styles on students' comprehension and memory recall.



HYPOTHESIS TO BE TESTED:

- Null Hypothesis (H_0): There is no significant effect of note-taking and learning styles on the comprehension and memory recall of students.
- Alternative Hypothesis (H_1): There is a significant effect of note-taking and learning styles on the comprehension and memory recall of students.

METHODOLOGY

DESIGN

2x3 Factorial Design

FACTORS AND LEVELS

- Factor A: Learning Styles
 - Level 1: Auditory
 - Level 2: Reading
 - Level 3: Visual
- Factor B: Presence of Note-Taking
 - Level 1: With Notes
 - Level 2: Without Notes

RESPONSE VARIABLE(S)

- Comprehension (short quiz, 5 items)
- Memory Recall (short quiz, 5 items)

SAMPLE AND REPLICATION

36 participants (12 per learning styles & 18 per presence of note-taking)

RANDOMIZATION

Participants were randomly assigned to learning style groups and note-taking conditions using pre-coded, shuffled consent forms.

DATA COLLECTION PROCEDURE

1. Pre-test & Post-test
2. Consent forms
3. Assignment
4. Passage exposure
5. Groups
6. Controlled conditions
7. Post-test
8. Data collected

RESULTS

Descriptive Statistics

Descriptive Statistics									
	N Statistic	Minimum Statistic	Maximum Statistic	Mean Statistic	Std. Deviation Statistic	Skewness		Kurtosis	
						Statistic	Std. Error	Statistic	Std. Error
PostTestK	36	2	5	3.92	.806	-.190	.393	-.630	.768
PostTestU	36	1	5	3.97	1.276	-.994	.393	-.188	.768
Valid N (listwise)	36								

Descriptive results show the scores for comprehension (PostTestK) and memory recall (PostTestU), with comprehension ranging from 2 to 5 ($M = 3.92$, $SD = 0.806$) and memory recall ranging from 1 to 5 ($M = 3.97$, $SD = 1.276$). The higher standard deviation in memory recall indicates greater variability in students' performance compared to comprehension.

ASSUMPTION 5

Test of Univariate Outliers

Descriptive Statistics									
	N Statistic	Minimum Statistic	Maximum Statistic	Mean Statistic	Std. Deviation Statistic	Skewness		Kurtosis	
						Statistic	Std. Error	Statistic	Std. Error
PostTestK	36	2	5	3.92	.806	-.190	.393	-.630	.768
PostTestU	36	1	5	3.97	1.276	-.994	.393	-.188	.768
Valid N (listwise)	36								

Descriptive results show both dependent variables; comprehension and memory recall – have acceptable skewness values (-0.190 and -0.994). No univariate outliers were detected, indicating no extreme scores.

ASSUMPTION 5

Test of Multivariate Outliers

Residuals Statistics ^a					
	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	16.07	22.85	18.50	1.945	36
Std. Predicted Value	-1.250	2.236	.000	1.000	36
Standard Error of Predicted Value	1.787	5.548	2.966	.838	36
Adjusted Predicted Value	14.44	23.96	18.73	2.324	36
Residual	-16.459	17.930	.000	10.355	36
Std. Residual	-1.543	1.681	.000	.971	36
Stud. Residual	-1.582	1.756	-.010	1.011	36
Deleted Residual	-17.296	19.565	-.234	11.236	36
Stud. Deleted Residual	-1.621	1.817	-.008	1.024	36
Mahal. Distance	.011	8.500	1.944	1.778	36
Cook's Distance	.000	.140	.029	.031	36
Centered Leverage Value	.000	.243	.056	.051	36

a. Dependent Variable: ParticipantNo

PValueMAH	
.9946	
.9946	
.9946	
.9946	
.7122	
.7122	
.7122	
.7122	
.7122	
.7122	
.6967	
.6967	
.6967	
.4820	
.4820	
.4793	
.4793	
.3769	
.3769	
.3769	
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.3769	
.3769	
.3769	
.3769	
.3667	
.2448	

.2448
.2431
.2431
.2391
.2391
.2391
.2391
.0615
.0432
.0143

Mahalanobis distance results show a minimum p-value of 0.0143, with all p-values above the 0.001 cutoff. This indicates that no multivariate outliers were present, and all participants fell within the acceptable range. Therefore, the dataset meets the assumption for multivariate analysis, with no extreme score combinations affecting the covariance structure.

ASSUMPTION 6

Test of Normality

Tests of Normality

	LearningStyle	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
PostTestK	Auditory	.257	12	.028	.807	12	.011
	Reading	.237	12	.061	.891	12	.123
	Visual	.279	12	.011	.784	12	.006
PostTestU	Auditory	.357	12	<.001	.710	12	.001
	Reading	.282	12	.009	.770	12	.004
	Visual	.238	12	.059	.798	12	.009

a. Lilliefors Significance Correction

1) The Shapiro-Wilk test for normality showed that most groups violated normality ($p < 0.05$), except for the Reading group's comprehension scores ($p = 0.123$). Both the With Notes and Without Notes groups also showed non-normality for comprehension and memory recall.

2) Despite these violations, MANOVA is generally robust to moderate deviations from normality, especially with balanced group sizes. Therefore, it was deemed appropriate to proceed with MANOVA without data transformation or non-parametric alternatives.

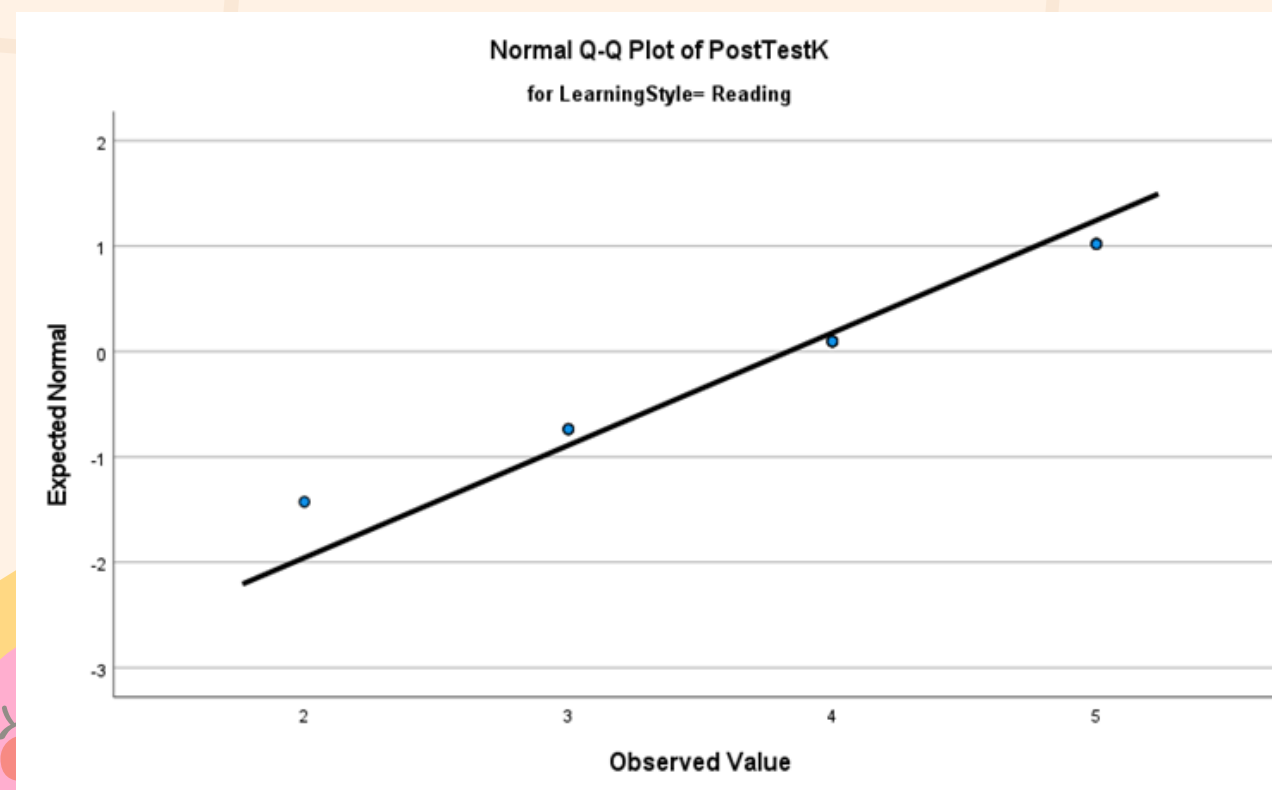
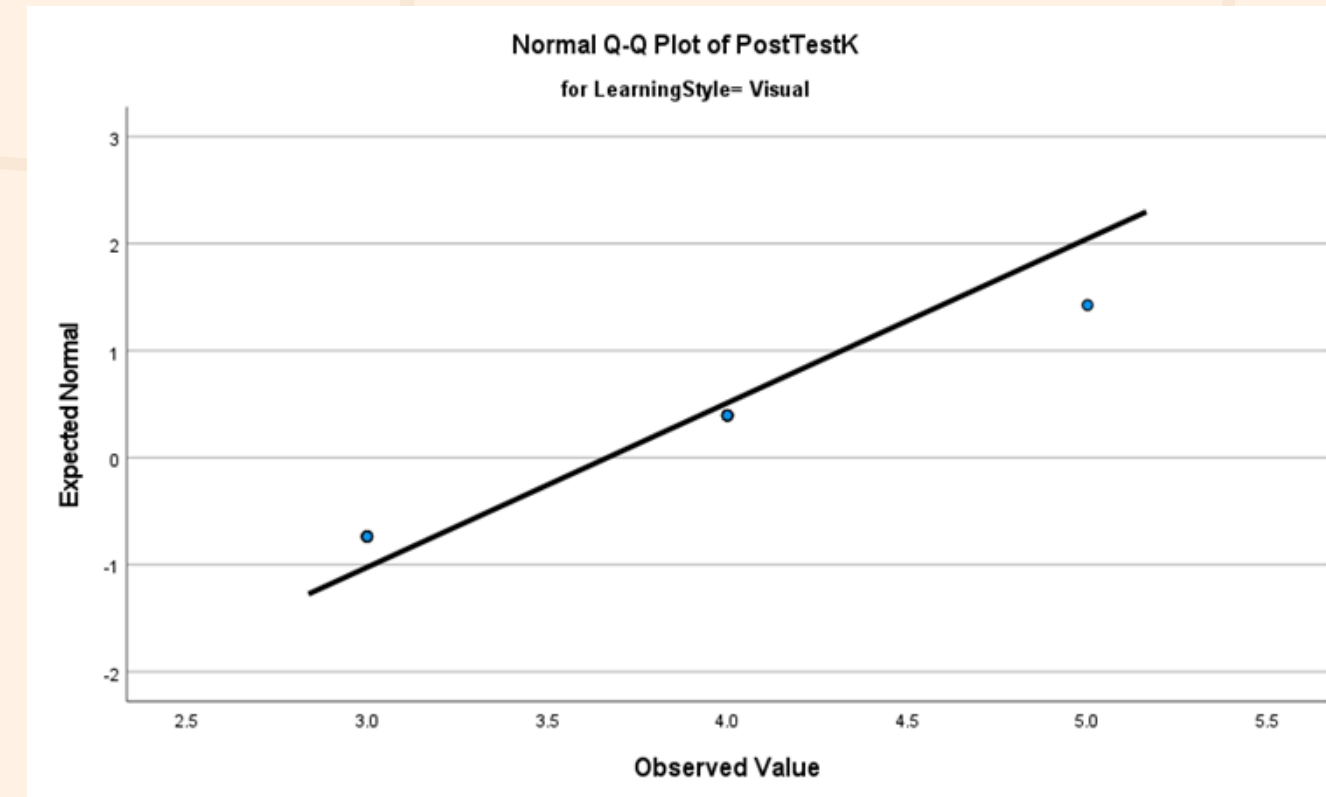
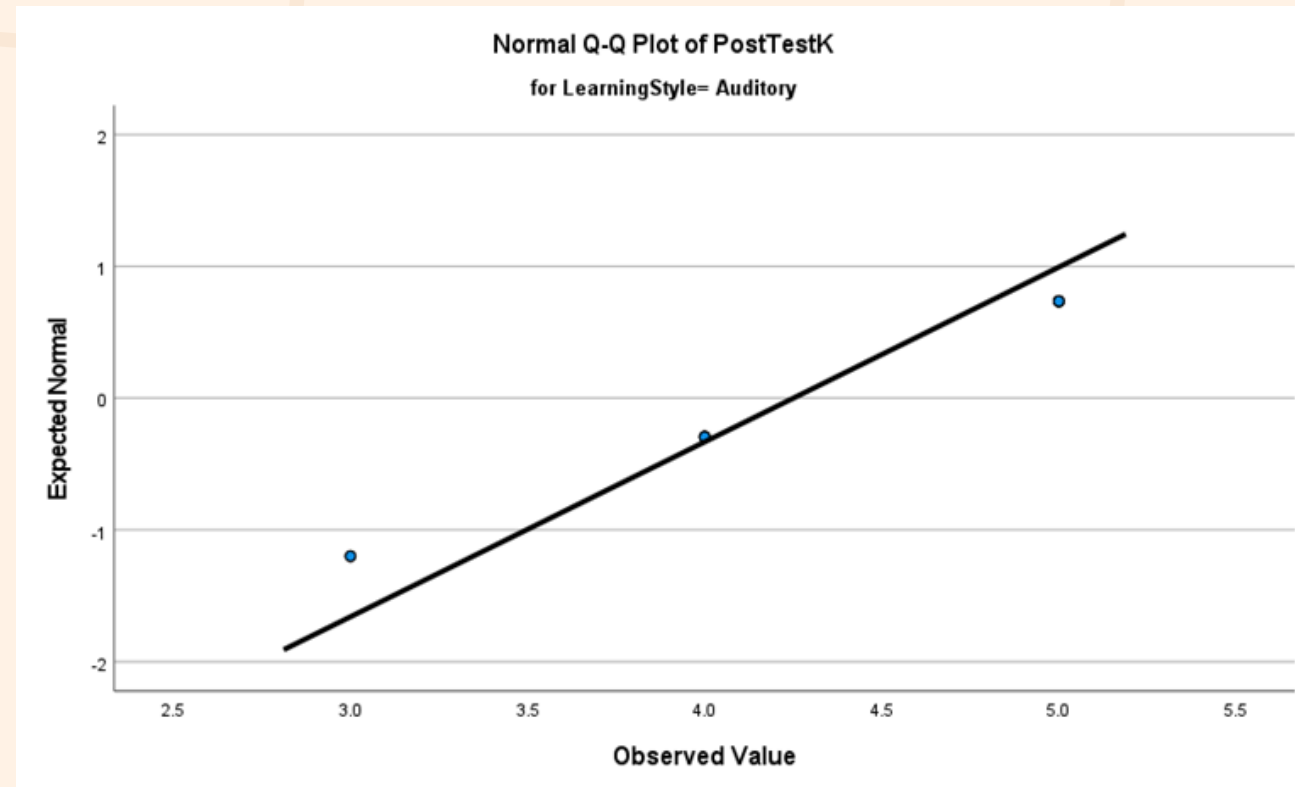
Tests of Normality

	WNorWON	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
PostTestK	With Notes	.244	16	.012	.878	16	.036
	Without Notes	.226	20	.009	.816	20	.002
PostTestU	With Notes	.384	16	<.001	.703	16	<.001
	Without Notes	.270	20	<.001	.807	20	.001

a. Lilliefors Significance Correction

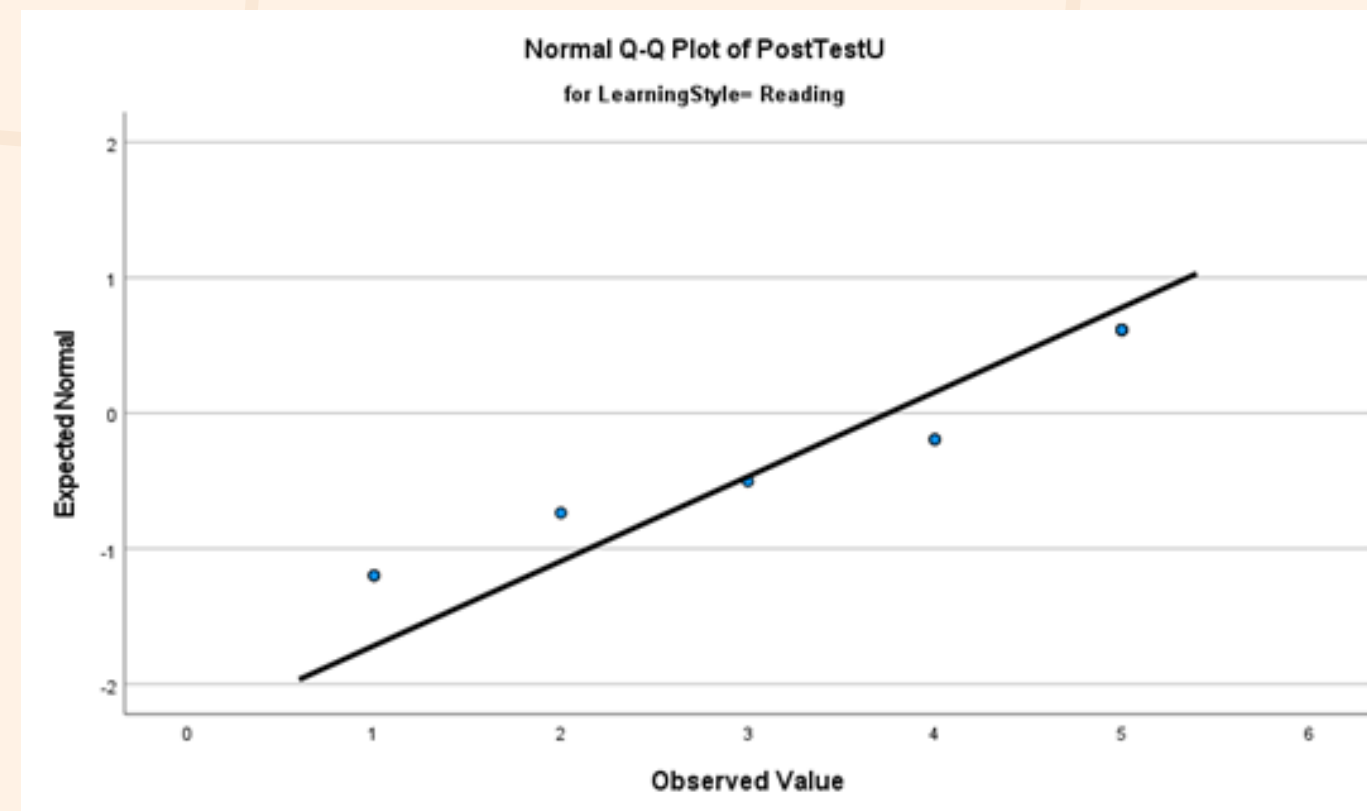
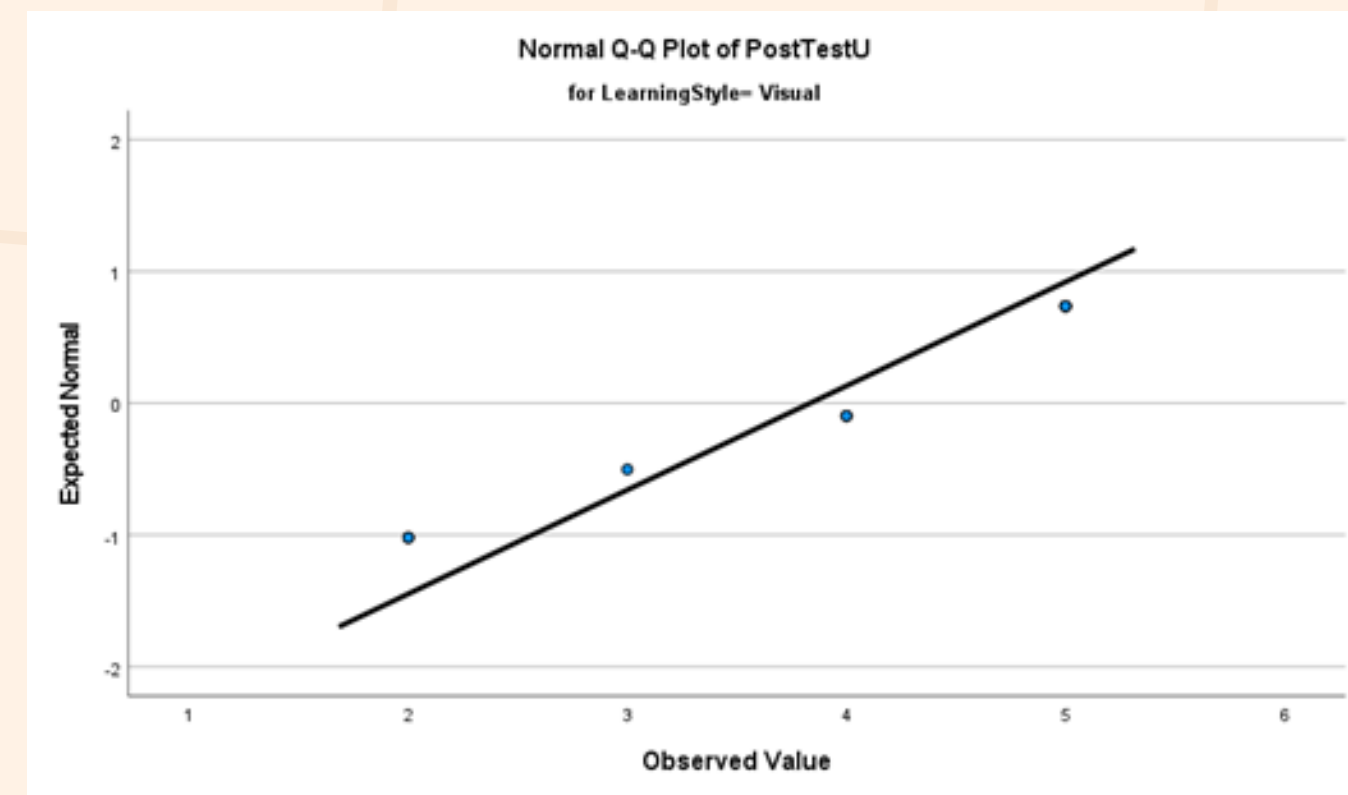
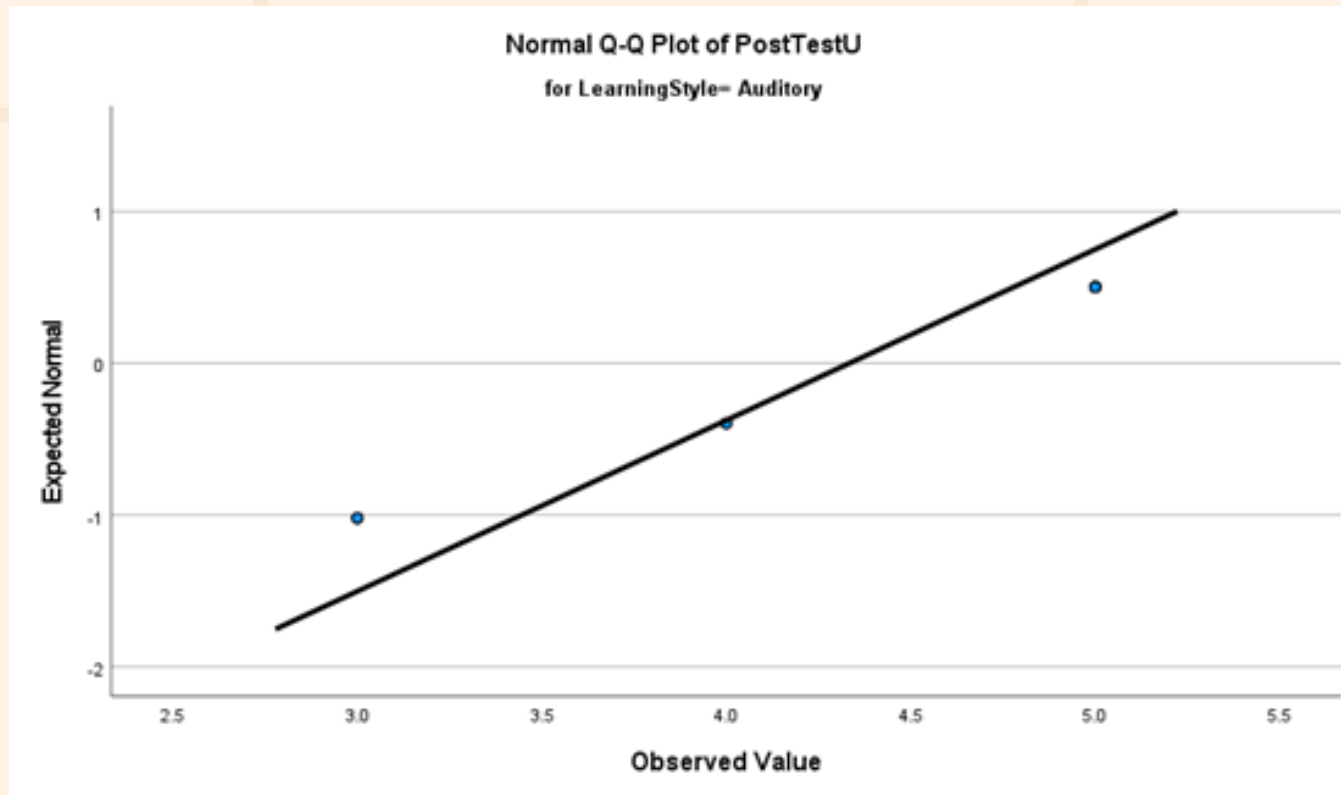
ASSUMPTION 7

Test of Linearity

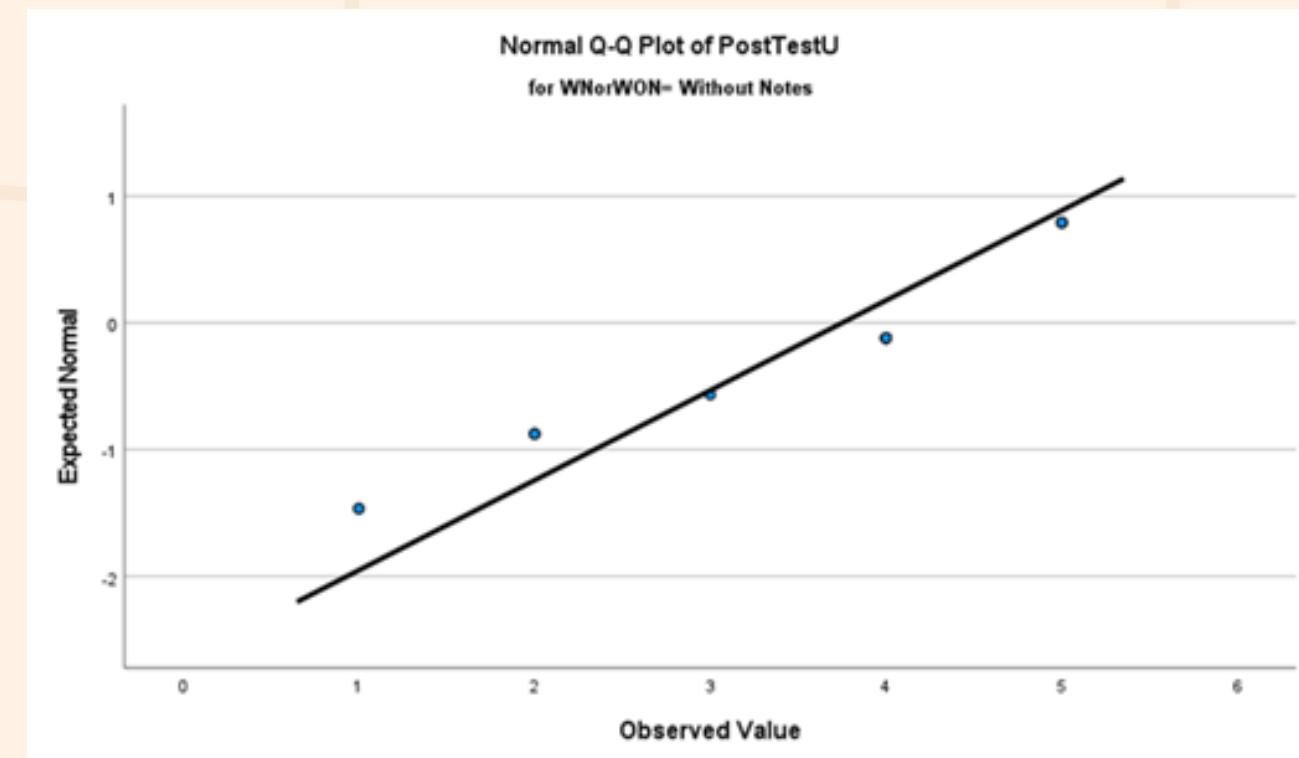
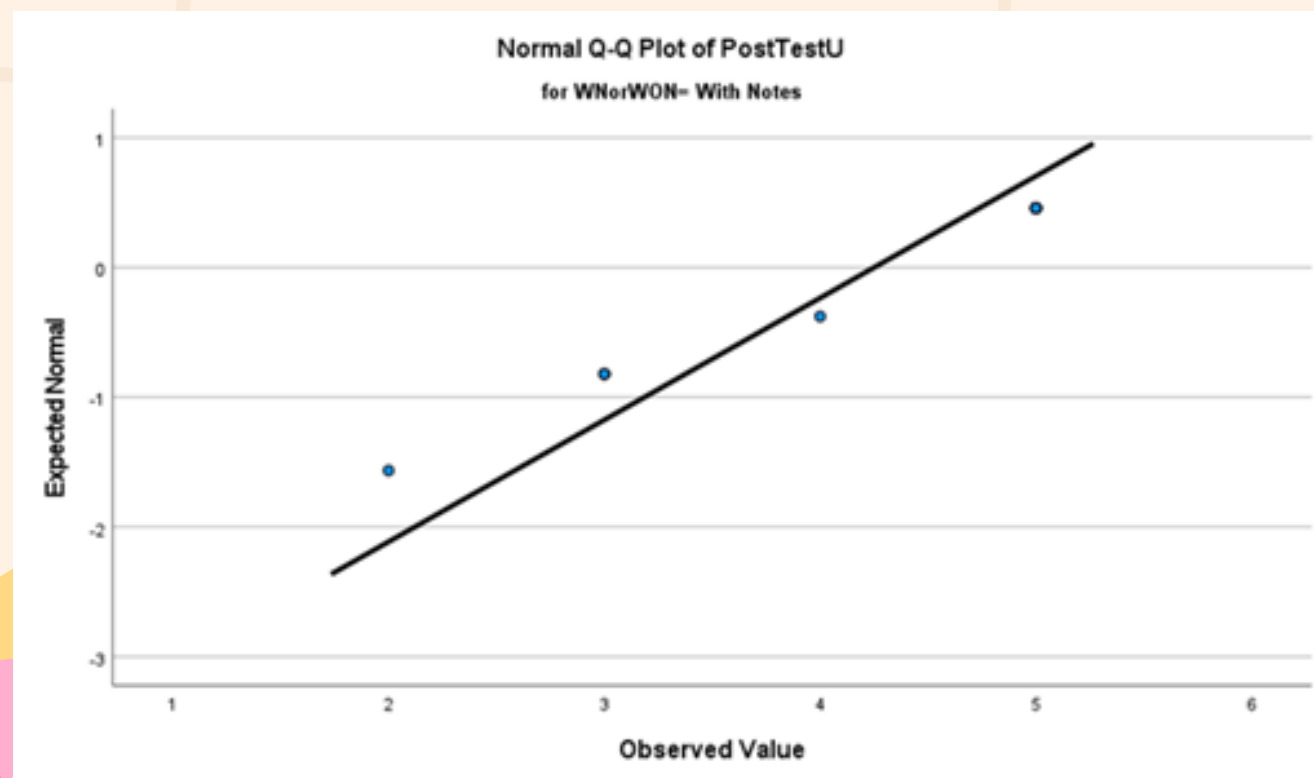
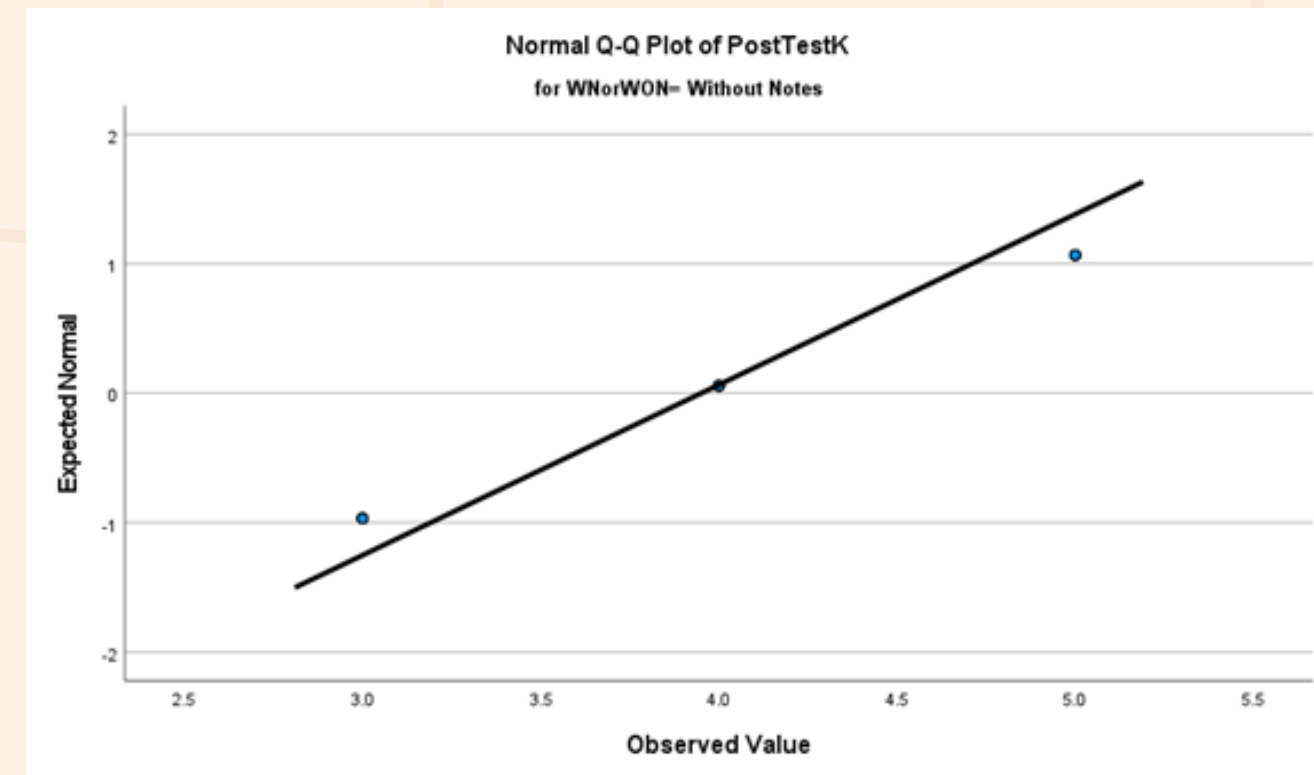
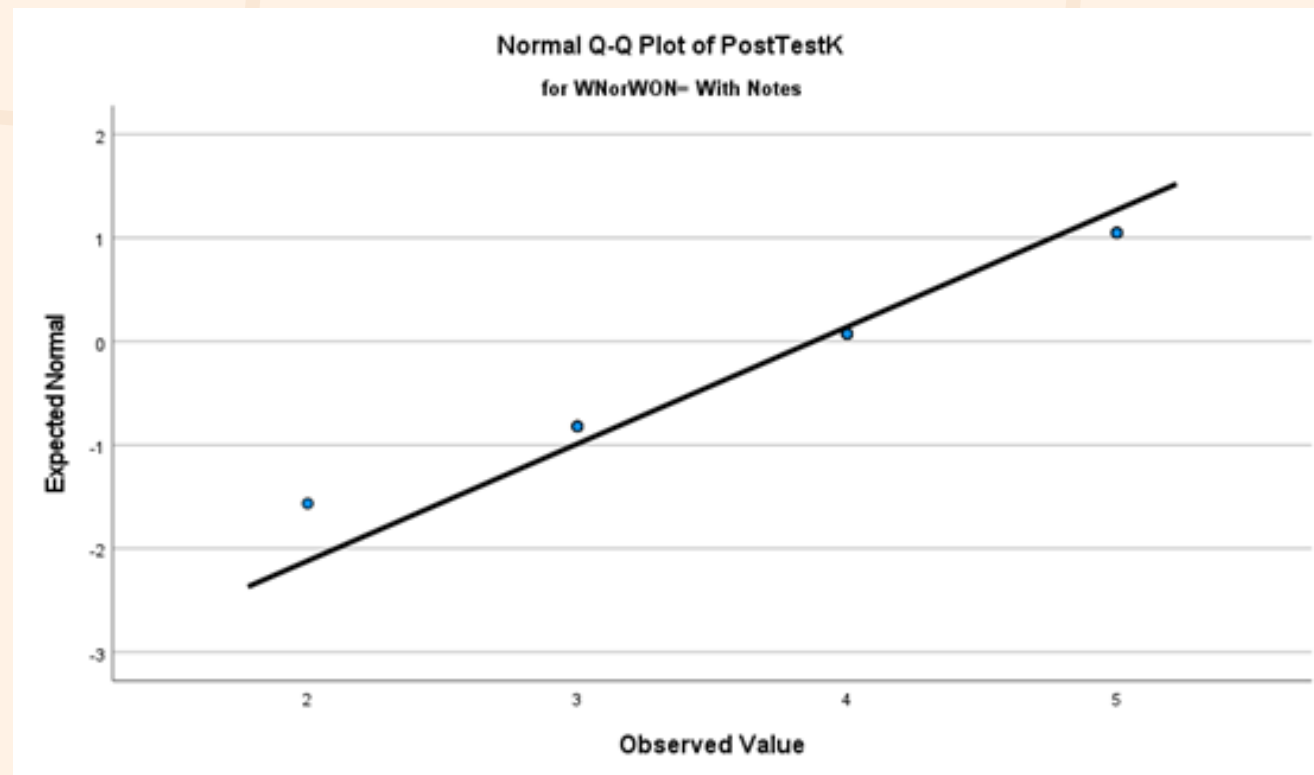


The Normal QQ plots show that the comprehension and memory recall scores mostly follow the diagonal lines, with only minor deviations at the tails, indicating approximate normality.

ASSUMPTION 7



ASSUMPTION 7



ASSUMPTION 8

Test of Homogeneity of Variance



Levene's Test of Equality of Error Variances^a

		Levene Statistic	df1	df2	Sig.
PostTestK	Based on Mean	.990	5	30	.441
	Based on Median	.943	5	30	.468
	Based on Median and with adjusted df	.943	5	19.896	.475
	Based on trimmed mean	.994	5	30	.438
PostTestU	Based on Mean	2.174	5	30	.084
	Based on Median	.725	5	30	.610
	Based on Median and with adjusted df	.725	5	24.797	.611
	Based on trimmed mean	2.018	5	30	.105

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Design: Intercept + LearningStyle + WNorWON + LearningStyle * WNorWON

Levene's Test results for comprehension (PostTestK) and memory recall (PostTestU) show all p-values greater than 0.05, indicating no significant differences in variances among groups. This confirms that the data meet the assumption of homogeneity of variance.



ASSUMPTION 8

Test of Homogeneity of Covariance

Box's Test of Equality of Covariance Matrices for comprehension (PostTestK) and memory recall (PostTestU) across learning styles and note-taking conditions showed Box's $M = 12.335$, $F = 0.670$, $p = 0.816$. Since the p -value is above 0.05, there is no significant difference in covariance matrices across groups.

Box's Test of Equality of Covariance Matrices^a

Box's M	12.335
F	.670
df1	15
df2	2618.490
Sig.	.816

Tests the null hypothesis that the observed covariance matrices of the dependent variables are equal across groups.

a. Design:
Intercept +
LearningStyle +
WNorWON +
LearningStyle *
WNorWON



ASSUMPTION 9

Test of Multicollinearity

Correlations

		PostTestK	PostTestU
PostTestK	Pearson Correlation	1	.331 [*]
	Sig. (2-tailed)		.049
	N	36	36
PostTestU	Pearson Correlation	.331 [*]	1
	Sig. (2-tailed)	.049	
	N	36	36

*. Correlation is significant at the 0.05 level (2-tailed).

The correlation between comprehension (PostTestK) and memory recall (PostTestU) is low ($r = 0.331$), indicating a weak relationship. This falls within the acceptable range, showing no multicollinearity, so both variables can be used together in the analysis without affecting the results.

TWO WAY MANOVA

Multivariate Tests ^a									
Effect		Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^d
Intercept	Pillai's Trace	.968	432.749 ^b	2.000	29.000	<.001	.968	865.499	1.000
	Wilks' Lambda	.032	432.749 ^b	2.000	29.000	<.001	.968	865.499	1.000
	Hotelling's Trace	29.845	432.749 ^b	2.000	29.000	<.001	.968	865.499	1.000
	Roy's Largest Root	29.845	432.749 ^b	2.000	29.000	<.001	.968	865.499	1.000
LearningStyle	Pillai's Trace	.114	.903	4.000	60.000	.468	.057	3.613	.270
	Wilks' Lambda	.886	.901 ^b	4.000	58.000	.469	.059	3.604	.269
	Hotelling's Trace	.128	.897	4.000	56.000	.472	.060	3.587	.267
	Roy's Largest Root	.128	1.919 ^c	2.000	30.000	.164	.113	3.839	.367
WNorWON	Pillai's Trace	.062	.966 ^b	2.000	29.000	.392	.062	1.932	.201
	Wilks' Lambda	.938	.966 ^b	2.000	29.000	.392	.062	1.932	.201
	Hotelling's Trace	.067	.966 ^b	2.000	29.000	.392	.062	1.932	.201
	Roy's Largest Root	.067	.966 ^b	2.000	29.000	.392	.062	1.932	.201
LearningStyle * WNorWON	Pillai's Trace	.160	1.300	4.000	60.000	.280	.080	5.201	.382
	Wilks' Lambda	.841	1.307 ^b	4.000	58.000	.278	.083	5.227	.383
	Hotelling's Trace	.187	1.310	4.000	56.000	.278	.086	5.240	.383
	Roy's Largest Root	.180	2.705 ^c	2.000	30.000	.083	.153	5.409	.495

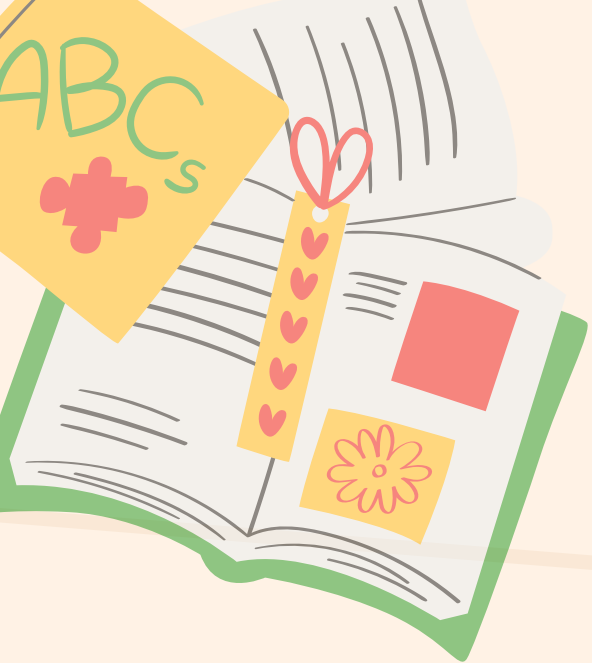
a. Design: Intercept + LearningStyle + WNorWON + LearningStyle * WNorWON

b. Exact statistic

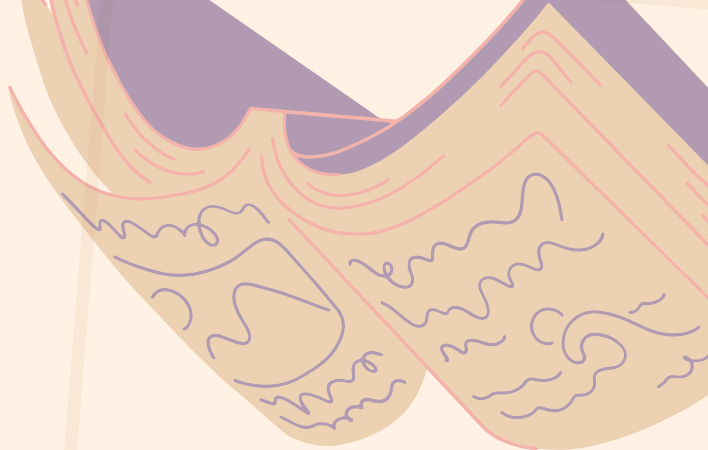
c. The statistic is an upper bound on F that yields a lower bound on the significance level.

d. Computed using alpha = .05

The Two-Way MANOVA results (Wilks' Lambda) show that learning style ($\Lambda = 0.886$, $F(4, 58) = 0.901$, $p = 0.469$, $\eta^2 = 0.059$) and note-taking ($\Lambda = 0.938$, $F(2, 29) = 0.966$, $p = 0.392$, $\eta^2 = 0.062$) both have very small effect sizes and do not significantly affect the combined dependent variables of comprehension (PostTestK) and memory recall (PostTestU).



TEST OF BETWEEN SUBJECT EFFECTS



Tests of Between-Subjects Effects							
Source	Dependent Variable	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	PostTestK	3.917 ^a	5	.783	1.248	.312	.172
	PostTestU	6.097 ^b	5	1.219	.719	.614	.107
Intercept	PostTestK	522.667	1	522.667	832.566	<.001	.965
	PostTestU	562.602	1	562.602	331.755	<.001	.917
LearningStyle	PostTestK	2.362	2	1.181	1.881	.170	.111
	PostTestU	1.657	2	.829	.489	.618	.032
WNorWON	PostTestK	.107	1	.107	.170	.683	.006
	PostTestU	2.042	1	2.042	1.204	.281	.039
LearningStyle * WNorWON	PostTestK	1.670	2	.835	1.330	.280	.081
	PostTestU	1.862	2	.931	.549	.583	.035
Error	PostTestK	18.833	30	.628			
	PostTestU	50.875	30	1.696			
Total	PostTestK	575.000	36				
	PostTestU	625.000	36				
Corrected Total	PostTestK	22.750	35				
	PostTestU	56.972	35				

a. R Squared = .172 (Adjusted R Squared = .034)
b. R Squared = .107 (Adjusted R Squared = -.042)

Test of between-subjects effects show that neither learning style nor note-taking had a statistically significant influence on comprehension or memory recall. Learning style showed non-significant results for comprehension, $F(2,30) = 1.861$, $p = 0.170$, partial eta-squared = 0.111 (medium effect), and for memory recall, $F(2,30) = 0.489$, $p = 0.618$, partial eta-squared = 0.032 (small effect), while note-taking also produced non-significant effects for comprehension, $F(1,30) = 0.170$, $p = 0.683$, partial eta-squared = 0.006, and memory recall, $F(1,30) = 1.204$, $p = 0.281$, partial eta-squared = 0.039. Overall, although some effect sizes fall within the small to medium range, these were not strong enough to reach statistical significance, indicating that students performed similarly regardless of learning style or note-taking condition.

DISCUSSION

1) Results:

- No significant differences in comprehension or recall based on note-taking or learning style
- No significant interaction between note-taking and learning styles
- Failed to reject null hypothesis

2) Possible Reasons for Non-Significant Results:

- Similar mean scores across groups
- Note-taking quality/method not controlled
- Learning styles may oversimplify individual differences
- Tasks may not have aligned strongly with learning style strengths

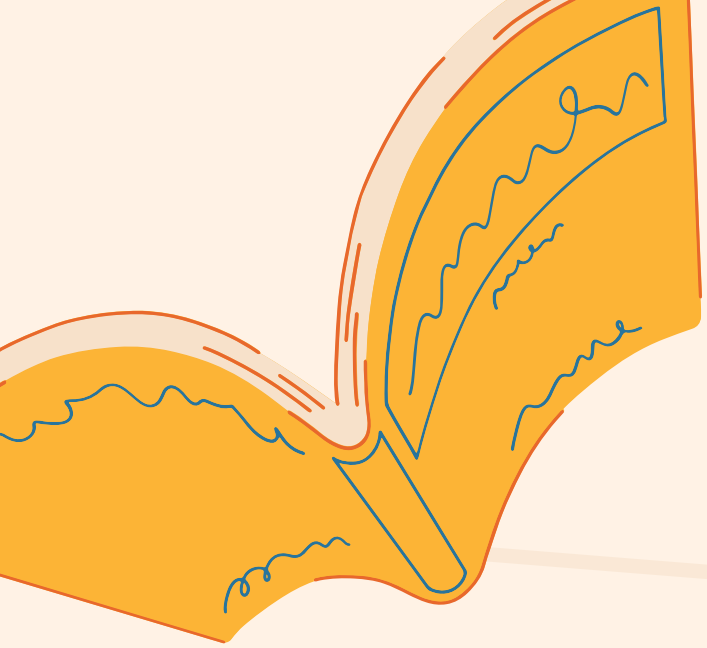
3) Limitations:

- Small sample size (36 respondents)
- Potential participant biases
- Uncontrolled variables (motivation, prior knowledge, environment)

4) Recommendations:

- Use larger sample sizes and better control of statistical assumptions to improve reliability
- Refine learning style classifications and standardize note-taking methods
- Minimize uncontrolled factors such as prior knowledge, motivation, and environmental distractions to better understand their impact on learning outcomes





THANK YOU!

**PRESENTED BY:
DATA THE EXPLORER**

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